PREDICTING

# Loan Default Status

## Using Random Forest Classification

DATA SCIENCE CAPSTONE

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WGU D606 — Data Science Capstone | EBN2 Task 3

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# About Me

PRESENTER

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**Background & Focus Areas:**

- Data Science & Machine Learning
- Financial Analytics & Risk Modeling
- Predictive Modeling & Classification

ABOUT THIS PRESENTATION

### Today's Goal

Share how a machine learning model can help lenders predict which borrowers are most likely to default on their loans — in plain, everyday terms.

### What We'll Cover

- The problem we set out to solve
- How we gathered and prepared the data
- What we found and what it means
- Practical recommendations for lenders

# The Problem — Why This Matters

*Every year, lenders lose billions of dollars when borrowers fail to repay their loans. The challenge? Identifying risky borrowers before a loan is approved.*

## The Risk

Loan defaults cause major financial losses for banks and credit unions, contributing to economic instability (CFPB, 2022).

## The Gap

Simple rules may miss interactions. This study did not compare its forest with an existing credit-scoring system.

## The Opportunity

This model combines 13 selected borrower and loan attributes to produce a default-risk score.

**Research question: How well do the selected borrower and loan attributes distinguish historical loan default status?**

# Research Question and Evaluation

## Ranking reference

ROC-AUC of 0.5 represents chance ranking. The test score describes ranking across thresholds on held-out loans.

## Practical evaluation

Use the confusion matrix to assess caught defaults, missed defaults and the number of loans sent for review.

## How We Defined 'Success'

We report predictive metrics on held-out data. The study did not perform a formal significance test or establish causal effects:

### Accuracy

What % of all loans did the model classify correctly?

### Precision

Of loans flagged as defaults, how many actually defaulted?

### Recall

Of all actual defaults, how many did the model catch?

### ROC-AUC

Overall ability to separate defaulters from non-defaulters (0.5 = random, 1.0 = perfect)

# The Data — What We Worked With

**148,670**

loan records

from WGU's
Loan Default Dataset

## 14 Key Variables Analyzed

- Gender
- Loan Limit
- Loan Purpose
- Business vs. Personal
- Region
- Age
- Credit Type
- Occupancy Type
- Loan Amount ($)
- Annual Income ($)
- Credit Score
- Debt-to-Income Ratio
- Property Value ($)
- **Default Status (Target)**

### Class Imbalance — A Major Challenge:

Most loans in the dataset were repaid. Only ~16% of loans in our cleaned dataset resulted in defaults — making it harder for the model to learn what default looks like.



# Summary of the Data Analysis Process

### 1 Import & Inspect

Loaded 148,670 loan records from Excel. Confirmed 34 columns and data types.

### 2 Select Features

Narrowed to 14 relevant columns: 8 categorical (demographics) + 5 financial + 1 target.

### 3 Clean the Data

Removed rows with missing values. Final dataset: 121,447 records (81.7% retained).

### 4 Encode Categories

Converted text categories (e.g., Region, Gender) into numerical format the model can read.

### 5 Split the Data

70% for training, 15% for validation, 15% for final testing — stratified to preserve class balance.

### 6 Train & Tune Model

Trained a Random Forest model, then optimized it using a grid search across 48 combinations.

# Our Approach — The Random Forest Model

## What Is a Random Forest?

The model builds 500 randomized decision trees from the training data. Each tree estimates default risk for a loan.

The forest averages those estimates into a score. A decision threshold turns that score into a default label.

## Why Random Forest?

### ✓ Handles Complexity

Detects non-obvious patterns that simple rules miss (e.g., high debt + low income together)

### ✓ Handles Imbalanced Data

We set class_weight='balanced' so the model pays extra attention to the rare default cases

### Reduces variance

Averaging trees can reduce variance. It does not guarantee that a forest will avoid overfitting.

### Supports interpretation

Individual-feature importance was not evaluated here.

# Outline of Findings — Final Model Results

*After training on 85,012 loans and testing on 18,218 loans the model had never seen:*

**84.7%**

**Accuracy**

Correctly classified 84.7% of all loans

**73.7%**

**ROC-AUC**

Far above the 50% baseline of random chance

**26.5%**

**Default Recall**

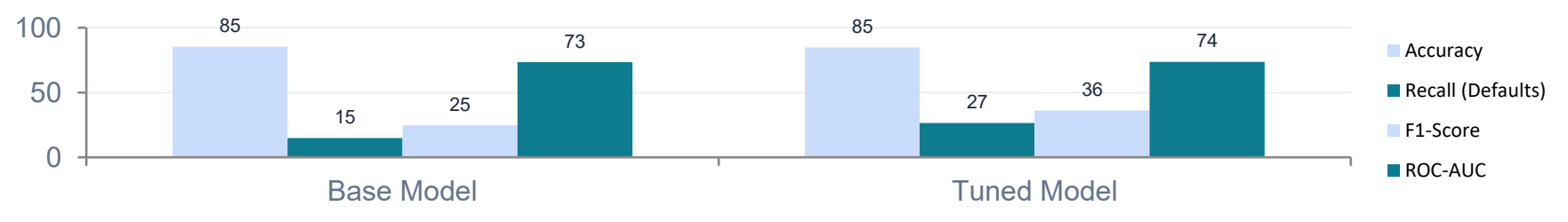
Correctly flagged 787 out of 2,973 defaults

**57.0%**

**Precision**

When it flagged a default, 57% were real defaults

**Baseline: validation | Tuned: test (different splits)**



# What the Model Learned — Key Insight

*The model ranked defaulted loans above repaid loans more often than chance in the held-out test set.*

### Test Set Results — Who Did the Model Catch?

| | Predicted: OK | Predicted: DEFAULT |
|---|---|---|
| **Actual: OK** | 14,652 ✓<br>Correctly cleared | 593 ✗<br>False alarms |
| **Actual: DEFAULT** | 2,186 ✗<br>Missed defaults | 787 ✓<br>Caught defaults |

**Held-out ranking**

**ROC-AUC 0.737**

The score summarizes ranking across thresholds. At the selected threshold, the model caught 26.5% of defaults.

*The biggest challenge: the model missed 2,186 defaults. Recall is the key metric to improve in future iterations (Lessmann et al., 2015).*

# Limitations of the Analysis

**01**

### Single Dataset — Limited Generalizability

The model was trained and tested on one institutional dataset from one time period. Results may not apply to other lenders, regions, or economic conditions. External validation was not performed.

**02**

### Missing Data Removed — Possible Selection Bias

We dropped 27,223 rows with missing values. If those borrowers had different characteristics than the rest, the model may not represent them fairly — a form of selection bias.

**03**

### Low Recall — Many Defaults Still Missed

Even after tuning, the model only caught 26.5% of actual defaults. Nearly 3 out of every 4 defaulters in the test set were not flagged, limiting practical utility as a standalone tool.

**04**

### Fairness & Bias Not Evaluated

The dataset contains sensitive variables (Gender, Region, Age). We did not measure whether the model's error rates differ across demographic groups — a critical ethical and regulatory gap (Barocas & Hardt, 2019).

**05**

### Black Box — Hard to Explain Individual Decisions

With 500 decision trees, it is difficult to explain to a loan applicant exactly why they were flagged as high-risk. Tools like SHAP values would be needed to improve transparency.

# Summary of Proposed Actions

Based on our findings, here is what we recommend for financial institutions:

**Step 1**

### Evaluate a supervised pilot

After external validation and subgroup checks, test whether a score helps prioritize manual review. Keep lending decisions with qualified underwriters.

**Step 2**

### Calibrate the Risk Threshold

Adjust how aggressively the model flags defaults based on the institution's risk tolerance. Lower the threshold → catch more defaults (higher recall) but review more applications manually.

**Step 3**

### Upgrade the Model

Compare against more powerful algorithms (XGBoost, LightGBM) which often outperform Random Forest on imbalanced data. Use precision-recall curves to find the optimal cut-off point.

**Step 4**

### Add Explainability & Fairness Auditing

Implement SHAP to inspect predictions and audit error rates across Gender, Region and age. These checks inform further review; they do not guarantee equitable outcomes.

# Expected Benefits of This Study

What the held-out test can tell an underwriting team

**787**
Defaults Caught

Of 2,973 defaults in the test set, 787 were flagged. Financial savings would need to be measured in deployment.

**73.7%**
ROC-AUC Score

The test ROC-AUC was 0.737. A score of 0.5 corresponds to chance ranking. This measures ranking across thresholds.

**~81%**
Data Preserved

Cleaning retained 121,447 complete records from 148,670. Dropping rows can change which borrowers the model represents.

## Broader Impact

The score could help prioritize manual review. Validation at another lender and checks for subgroup performance would be needed before making underwriting decisions.

# References

*All sources are cited in APA format. In-text citations appear throughout the presentation where content is paraphrased or referenced.*


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# Thank You

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*Test ROC-AUC: 0.737 | Default recall: 26.5% | Potential use: manual review prioritization*
*Financial benefits remain unmeasured.*